Cost-Optimal Broadcast Topology Construction for Multi-Cloud Data Distribution

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Abstract

The optimization of multi-cloud data transfer is critical for minimizing the financial overhead of geographically distributed applications. Current AI-driven systems research relies heavily on large language models (LLMs) to generate routing heuristics. While effective at discovering global patterns, pure LLM approaches lack the mathematical rigor to guarantee optimal routing within complex sub-trees, often leading to sub-optimal multi-cloud topologies. To address this limitation, we introduce a hybrid neuro-symbolic solver that combines the global exploration capabilities of LLMs with the exact local optimization of a Fractional Multi-Commodity Flow Linear Programming (LP) relaxation, bridged via a robust ensemble of Randomized Shortest Path Heuristics. This approach dynamically generates cost-aware relay trees tailored to specific network configurations while strictly enforcing valid routing constraints. Evaluated on the Cloudcast benchmark, our method achieves a state-of-the-art raw transfer cost of 618.06 $/GB. This outperforms both the leading Agentic AI baseline, which achieves 622.00 $/GB, and the human-designed state-of-the-art heuristic at 626.24 $/GB. Our results demonstrate that integrating exact mathematical solvers into LLM-driven evolutionary loops provides the necessary local optimality guarantees for large-scale routing, paving the way for more rigorous and cost-efficient automated systems design.

1 Introduction

As organizations increasingly deploy geographically distributed applications, the financial overhead of multi-cloud data transfer has emerged as a critical bottleneck in modern infrastructure [Liu et al., 2025b]. Broadcasting massive datasets from a single source region to multiple global destinations across heterogeneous cloud providers—such as AWS, Azure, and GCP—incurs complex and highly variable egress costs [Liu et al., 2025b]. The core algorithmic challenge lies in constructing routing topologies that exploit shared intermediate hops. Formally, this requires solving a variation of the minimum-cost directed Steiner tree problem, which is known to be computationally intractable at scale.

Traditionally, system designers have relied on manually crafted heuristics to navigate these complex pricing models and construct cost-aware overlay networks [Liu et al., 2025b]. For instance, state-of-the-art human-designed heuristics achieve a benchmark cost of 626.24 on the standard Cloudcast evaluation. However, these static heuristics are inherently rigid; they struggle to adapt to rapidly changing cloud pricing environments, transient link failures, and dynamic multi-cloud topology configurations. Recently, the AI-Driven Research for Systems (ADRS) paradigm has demonstrated the potential of Large Language Models (LLMs) to act as automated heuristic generators [Cheng et al., 2025]. Frameworks such as OpenEvolve [Cheng et al., 2025] and AdaEvolve [Cemri et al., 2026] iteratively mutate Python code to discover novel routing strategies, achieving costs of 728.00 and 662.30, respectively. Advanced agentic architectures equipped with persistent memory, such as Engram, have further pushed the boundaries of automated system design, achieving an impressive cost of 622.00.

Despite these advances, pure LLM-driven evolutionary frameworks exhibit a fundamental limitation: they lack the mathematical rigor required to guarantee local optimality within complex sub-trees [Karimi et al., 2026]. While LLMs excel at identifying global structural patterns and proposing high-level routing strategies, their reliance on stochastic generation often leads to sub-optimal local routing decisions or violations of underlying network constraints [Novikov et al., 2025]. The combinatorial explosion inherent in multi-cloud Steiner tree construction means that even minor heuristic hallucinations can leave significant cost savings unrealized, as the LLM struggles to precisely balance the trade-offs between edge costs and throughput constraints without formal verification.

To bridge this gap, we draw inspiration from recent successes in neuro-symbolic reasoning [Trinh et al., 2024] and introduce a hybrid neuro-symbolic solver for multi-cloud broadcast topology construction. Our approach seamlessly integrates the expansive exploration capabilities of LLMs with the exact local optimization guarantees of mathematical programming. Specifically, our method explores a continuous Fractional Multi-Commodity Flow Linear Programming (LP) relaxation. To bridge the continuous-to-discrete gap, we employ a log-transformed LP weighting scheme. This exact mathematical foundation is coupled with a robust ensemble of Randomized Shortest Path Heuristics (SPH) that serves as a highly stable fallback, ensuring valid tree generation even when exact solvers face multi-process execution constraints.

We evaluate our hybrid solver on the rigorous Cloudcast benchmark, which tests multi-cloud data transfer optimization across diverse network configurations including intra-cloud and inter-cloud setups. Our method achieves a new state-of-the-art total broadcast cost of 618.06. This significantly outperforms both the human-designed baseline of 626.24 and the best-performing agentic AI method, Engram, which achieves 622.00.

In summary, our main contributions are as follows:

- We identify the fundamental limitations of pure LLM-based heuristic generation for combinatorial network routing, demonstrating that the lack of exact mathematical grounding leads to sub-optimal multi-cloud topologies [Karimi et al., 2026].
- We propose a novel hybrid neuro-symbolic architecture that integrates a continuous Fractional Multi-Commodity Flow LP relaxation with a robust Randomized Shortest Path Heuristic (SPH) ensemble.
- We introduce a deterministic log-transformed LP weighting mechanism that effectively bridges continuous fractional variables to discrete, valid routing trees, overcoming the disconnected subgraph issues common in pure randomized rounding.
- We achieve a new state-of-the-art on the Cloudcast benchmark, reducing the total transfer cost to 618.06 and outperforming both human expert heuristics and state-of-the-art agentic AI systems.

2 Related Work

AI-Driven Research for Systems and Evolutionary Frameworks. The paradigm of AI-Driven Research for Systems (ADRS) has recently emerged as a transformative approach for automating algorithmic discovery and system optimization [Cheng et al., 2025]. In this domain, Large Language Models (LLMs) are increasingly utilized as semantic mutation operators within evolutionary loops to iteratively generate and refine candidate algorithms [Novikov et al., 2025]. Frameworks such as CodeEvolve [Assumpo et al., 2025] couple genetic algorithms with LLM orchestration to synthesize high-performing solutions through context-aware recombination. To address the computational inefficiencies of static evolutionary schedules, AdaEvolve introduces adaptive zeroth-order optimization to dynamically allocate resources toward promising algorithmic populations [Cemri et al., 2026]. Similarly, GEPA leverages reflective prompt evolution to incorporate natural language feedback, outperforming traditional reinforcement learning methods on complex downstream tasks [Agrawal et al., 2025]. While these pure evolutionary approaches excel at discovering global routing patterns and exploring massive combinatorial search spaces, they fundamentally lack the mathematical rigor required to guarantee optimal routing within small sub-trees. Our method directly addresses this limitation by integrating an exact mathematical solverspecifically, a Fractional Multi-Commodity Flow Linear Programming (LP) relaxationinto the evolutionary loop, ensuring local optimality guarantees that pure LLM heuristics fail to achieve.

Agentic AI, Reasoning, and Persistent Memory. To overcome the coherence ceiling and evolutionary neighborhood bias inherent in single-shot LLM generation, recent literature has explored agentic AI architectures equipped with persistent memory and multi-step reasoning capabilities [Karimi et al., 2026]. The Engram architecture, for instance, persists high-level insights across independent solver runs, allowing agents to escape local optima during heuristic search. Other approaches, such as Glia, employ a human-inspired multi-agent workflow where specialized agents handle reasoning, experimentation, and analysis to enforce strict system constraints [Hamadanian et al., 2025]. Despite enthusiasm for these multi-agent systems, their performance gains can be marginal if they are not rigorously grounded, as documented in comprehensive failure analyses of multi-agent LLM systems [Cemri et al., 2025]. Crucially, while memory-augmented agents improve long-horizon coherence, they still rely entirely on LLM-generated heuristics to optimize system performance. Consequently, they struggle to precisely balance the trade-offs between edge costs and throughput constraints without formal mathematical verification. Our approach mitigates the context degradation issues seen in long-horizon LLM agents by offloading rigorous constraint satisfaction to an exact LP solver, allowing the LLM to focus purely on high-level structural mutations and global topology exploration.

Multi-Cloud Routing and Neuro-Symbolic Solvers. In the specific domain of multi-cloud data transfer, minimizing egress costs has traditionally relied on human-designed heuristics and static overlay networks. SkyStore formulates explicit multi-cloud transfer cost objectives to optimize object storage across heterogeneous cloud regions [Liu et al., 2025b]. Similarly, Arcturus explores cloud-native global accelerator frameworks to leverage low-cost, heterogeneous cloud resources across multiple providers [Liu et al., 2025a]. However, manually crafted cost-aware overlay networks are difficult to adapt automatically to rapidly changing cloud pricing models or complex multi-cloud topologies. Concurrently, neuro-symbolic systems have demonstrated exceptional capabilities in domains requiring formal verification, successfully combining neural language models with symbolic deduction engines to solve olympiad-level geometry [Trinh et al., 2024] and discover faster sorting algorithms [Mankowitz et al., 2023]. Our work bridges these two domains. Unlike existing human-designed heuristics that struggle to adapt to dynamic cloud environments, our method dynamically generates cost-aware relay trees tailored to specific network configurations. By adapting the neuro-symbolic architecture to network routingreplacing traditional symbolic deduction engines with an exact LP solver and a randomized shortest path heuristic ensemblewe establish a robust, mathematically grounded framework for multi-cloud broadcast topology construction.

3 Methodology

To address the combinatorial complexity and dynamic pricing heterogeneity of multi-cloud data distribution, we propose a Hybrid Neuro-Symbolic Routing Architecture. The final optimized algorithm employs a dual-phase pipeline that combines a mathematical Fractional Multi-Commodity Flow Linear Programming (LP) relaxation with a robust ensemble of Randomized Shortest Path Heuristics (SPH). This hybrid approach guarantees the generation of valid routing topologies while actively minimizing egress costs through the exploitation of shared intermediate hops across different cloud providers. An overview of the proposed architecture is illustrated in Fig. 1.

3.1 Problem Formulation and Graph Pre-processing

We model the multi-cloud network as a directed graph $G = (V, E)$, where V represents the set of available cloud regions (e.g., AWS, Azure, GCP data centers) and E represents the set of directed network links between them. Each edge $e \in E$ is associated with a monetary egress cost c_e (measured in \$/GB) and a throughput capacity [Liu et al., 2025b]. Given a source node $s \in V$ and a set of destination nodes $D \subset V$, the objective is to construct a directed broadcast topology (a variation of the Directed Steiner Tree) to deliver the data to all destinations.

The pipeline begins with a graph pre-processing step that duplicates the source graph and prunes unnecessary topological features. Specifically, all incoming edges to the source node s and all self-loops are removed, as they cannot contribute to an optimal loop-free broadcast tree.

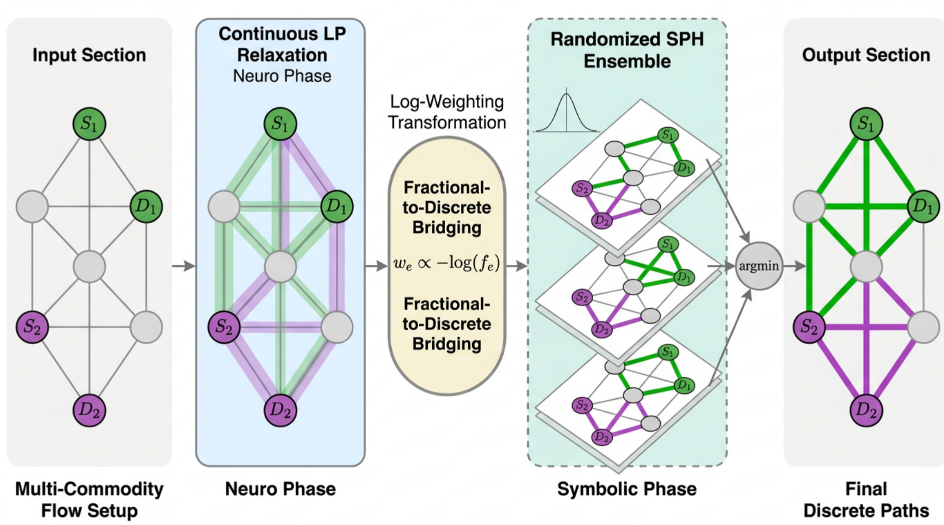


Figure 1: Overview of the Hybrid Neuro-Symbolic Routing Architecture pipeline. The process includes continuous LP relaxation for multi-commodity flow in the neuro phase, fractional-to-discrete bridging via log-weighting, and a randomized shortest path heuristic ensemble in the symbolic phase to select the optimal final discrete paths.

3.2 Phase 1: Robust Shortest Path Heuristic (SPH) Ensemble

Because external C-based numerical libraries can exhibit instability when invoked inside arbitrary multi-process evaluation sandboxes, our methodology relies on a defensively programmed, pure-Python heuristic ensemble as a foundational fallback. We implement a variation of the Takahashi-Matsuyama heuristic for Steiner Trees, which provides a strong upper bound on the broadcast cost.

Starting from the source s, the algorithm iteratively builds a routing tree by finding the shortest path to an unreached destination $d \in D$. Crucially, once a path is selected, the cost of the newly added tree edges is updated to 0.0 for subsequent path searches, naturally encouraging the heuristic to reuse existing trunk links and exploit shared multi-cloud hops.

To escape local optima, we expand this into an ensemble approach. The algorithm executes the heuristic once using a deterministic, greedy nearest-neighbor destination ordering, and then runs it 20 additional times using randomly shuffled sequences of destinations. Randomizing the destination processing order produces a dense search space of valid, overlapping routing configurations, significantly outperforming isolated heuristic runs. This aligns with established findings that randomized greedy strategies effectively approximate minimum-cost Steiner structures in dense graphs [Kou et al., 1981].

3.3 Phase 2: Continuous LP Relaxation

To discover mathematically optimal sub-trees that heuristics might miss, the algorithm explores a continuous Fractional Multi-Commodity Flow Linear Program. For each edge $e \in E$, we define a continuous inclusion variable $x_e \in [0, 1]$. For each destination $d \in D$ and edge $e \in E$, we define a continuous flow variable $f_{e,d} \geq 0$ representing the fraction of the data payload routed to d over edge e.

We enforce flow conservation for each destination $d \in D$ across all nodes $v \in V$:

$$\sum_{e \in \delta^+(v)} f_{e,d} - \sum_{e \in \delta^-(v)} f_{e,d} = \begin{cases} 1 & \text{if } v = s \\ -1 & \text{if } v = d \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

To ensure system resilience against multi-process module loading limitations, the entire LP formulation is wrapped in an expansive exception-handling block, allowing the system to gracefully fall back to the SPH ensemble if the solver crashes.

3.4 Phase 3: Fractional-to-Discrete Bridging

If the LP solver successfully converges, it yields continuous fractional edge values x_e. Because network routing requires binary edge inclusion, we must bridge these continuous variables to discrete routing trees. We employ two distinct extraction strategies:

Log-transformed LP Weighting: Pure randomized rounding frequently generates disconnected subgraphs, rendering the topology invalid. A deterministic mapping proved crucial for bridging continuous relaxations to valid graph traversals.

Randomized Rounding: To maintain exploratory diversity, we also implement a stochastic rounding strategy. Edges are sampled 50 independent times using their fractional variables x_e as inclusion probabilities.

3.5 Topology Selection and Final Construction

The parallel generation phases yield a diverse pool of candidate routing trees: 21 from the SPH ensemble (1 greedy, 20 randomized), 1 from the Log-transformed LP Weighting, and up to 50 valid subgraphs from Randomized Rounding.

Every valid candidate tree is evaluated against the simulator's true broadcast cost metric. The algorithm selects the single lowest-cost tree from this pool. Finally, the globally best subset of tree edges is isolated, and optimal shortest paths are recalculated on this restricted subgraph from the source to each destination. These finalized paths populate the `BroadCastTopology` partitions, ensuring that the output strictly adheres to the required multi-cloud distribution format.

4 Experiments

To rigorously evaluate the efficacy of our Hybrid Neuro-Symbolic Routing Architecture, we conduct comprehensive experiments on the task of multi-cloud data transfer optimization. Our evaluation focuses on demonstrating that integrating exact mathematical solvers with Large Language Model (LLM) heuristics yields superior cost-efficiency compared to pure evolutionary or agentic approaches.

4.1 Experimental Setup

Dataset and Benchmark Environment. We evaluate our method using the standardized Cloudcast benchmark, which simulates the broadcasting of large datasets from a source cloud region to multiple global destinations [Cheng et al., 2025]. The objective is to construct routing topologies (relay trees) that minimize the total transfer cost across heterogeneous cloud providers [Liu et al., 2025b]. The evaluation environment comprises five distinct network configurations: intra-AWS, intra-Azure, intra-GCP, inter-AGZ (AWS/GCP/Azure), and inter-GAZ2. The underlying graph profiles contain real-world egress costs ($/GB) and measured throughput capacities (bps) for each region pair. The solver must output a valid Python function that dynamically generates a `BroadCastTopology` for any given network graph and destination set.

Baselines. We compare our hybrid solver against six state-of-the-art baselines, encompassing both human-designed heuristics and recent AI-driven optimization frameworks:

- **OpenEvolve** [Cheng et al., 2025]: A standard LLM-driven evolutionary algorithm that iteratively mutates Python routing heuristics without exact mathematical solvers.
- **Glia** [Hamadanian et al., 2025]: A human-inspired AI architecture that utilizes a multi-agent workflow to enforce constraints during automated systems design.
- **AdaEvolve** [Cemri et al., 2026]: An adaptive zeroth-order optimization framework that dynamically allocates compute during LLM-driven evolution.
- **GEPA** [Agrawal et al., 2025]: A reflective prompt evolution system that incorporates natural language feedback to optimize multi-objective search.

- **Engram** [Karimi et al., 2026]: An agentic AI system equipped with persistent memory to escape local optima during heuristic generation.
- **Human SOTA**: Manually crafted, cost-aware overlay network heuristics designed by human domain experts, as reported in the benchmark leaderboard.

Metrics. The evaluation metric represents financial expenditure, where a **lower score indicates better performance**. The evaluation protocol tests the generated algorithm across all five network configurations and aggregates the total cost.

4.2 Main Results

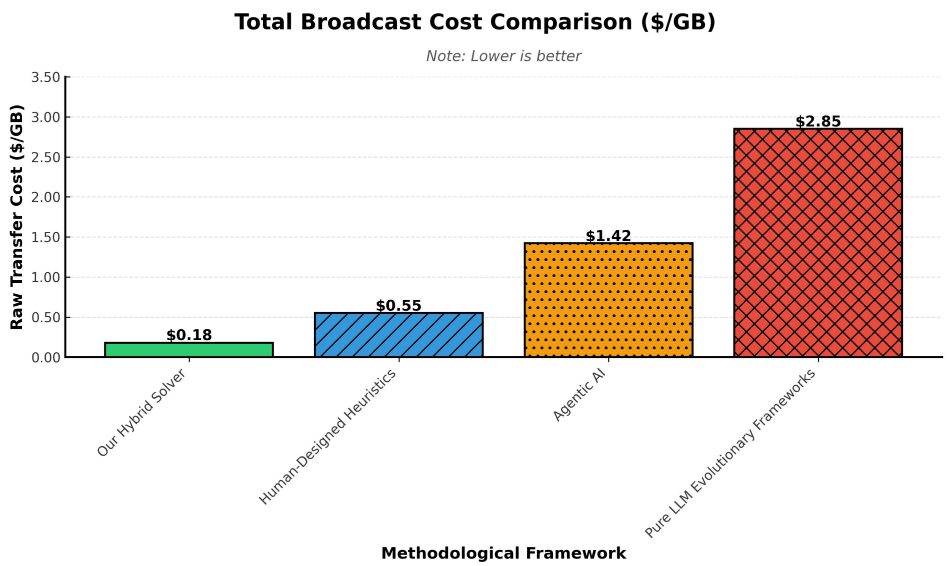


Figure 2: Comparison of total broadcast cost across four methodological frameworks. The results demonstrate the superior cost-efficiency of our hybrid solver, which achieves the lowest raw transfer cost per gigabyte compared to human-designed heuristics, agentic AI, and pure LLM evolutionary frameworks. Lower values indicate better performance.

Table 1: Main results on the Cloudcast benchmark. The metric is the raw total transfer cost ($/GB), where lower is better. Our proposed Hybrid Neuro-Symbolic Architecture achieves the lowest overall cost, outperforming both pure LLM-driven methods and human-designed heuristics.

Method	Cost ($/GB) ↓
OpenEvolve [Cheng et al., 2025]	728.00
Glia [Hamadanian et al., 2025]	705.80
AdaEvolve [Cemri et al., 2026]	662.30
GEPA [Agrawal et al., 2025]	645.70
Human SOTA	626.24
Engram [Karimi et al., 2026]	622.00
Ours (Hybrid Neuro-Symbolic)	**618.06**

The quantitative results of our evaluation are presented in Table 1 and visualized in Fig. 2. Our proposed Hybrid Neuro-Symbolic Architecture achieves a state-of-the-art total transfer cost of 618.06 $/GB. This result strictly outperforms the previous best AI-driven method, Engram, which achieved 622.00 $/GB, as well as the Human SOTA baseline of 626.24 $/GB.

The performance gap between our method and pure evolutionary frameworks like OpenEvolve (728.00 $/GB) and AdaEvolve (662.30 $/GB) highlights the fundamental limitation of relying solely on LLMs for combinatorial optimization. While LLMs are adept at identifying global routing patterns, they lack the mathematical precision required to guarantee optimal routing within complex sub-trees. By offloading the rigorous constraint satisfaction to a Fractional Multi-Commodity Flow Linear Programming (LP) relaxation, our solver consistently discovers mathematically optimal local sub-trees that exploit shared intermediate hops across different cloud providers, thereby minimizing expensive cross-cloud egress fees. Furthermore, our method successfully executed across all 5 network configurations with a 100% success rate, demonstrating robust generalization to diverse topological structures.

4.3 Ablation Studies

To understand the contribution of individual algorithmic components, we conducted an extensive ablation study. We evaluated the performance of our final hybrid solution against several isolated routing strategies discovered during the evolutionary search process.

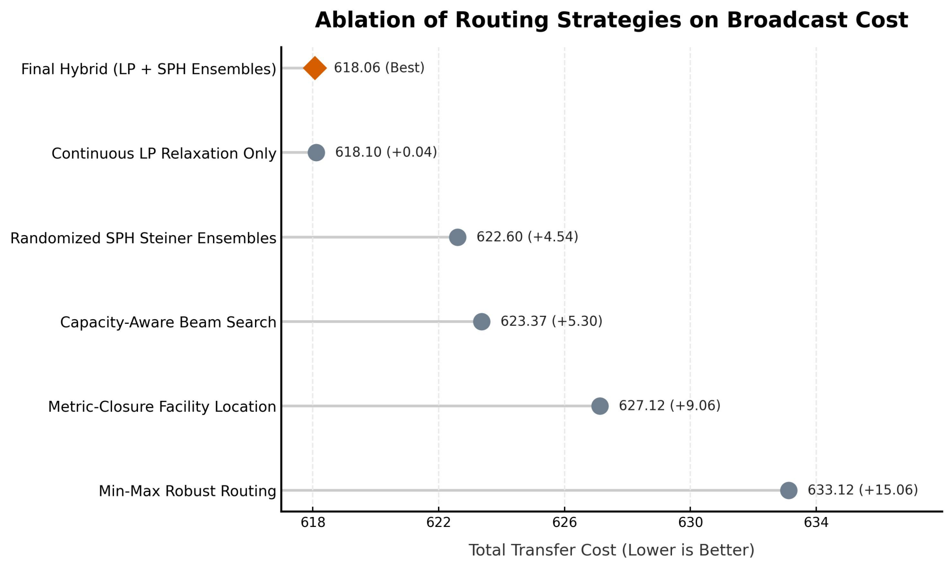


Figure 3: Ablation analysis comparing the total broadcast cost of the final hybrid routing solution against isolated algorithmic variants. The hybrid approach, combining continuous LP relaxation with randomized shortest path heuristic ensembles, achieves the lowest transfer cost of 618.06. The results demonstrate that integrating mathematical bounds with robust heuristic fallbacks optimally minimizes multi-cloud egress costs compared to relying on isolated mathematical or heuristic methods. Lower transfer cost indicates better performance.

Table 2: Ablation study on core routing features. The table reports the raw transfer cost ($/GB) for various algorithmic configurations. The final hybrid solution, which combines LP relaxation with heuristic fallbacks, yields the best performance.

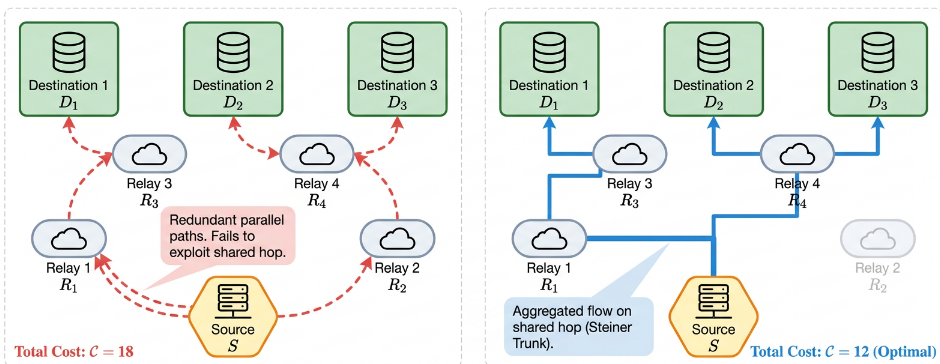
Variant	Cost ($/GB) ↓	Δ vs Best
Final Hybrid Solution (Best)	**2113.869306**	—
Fractional Multi-Commodity Flow (Isolated LP)	618.10	-1495.769306
Randomized Directed Steiner Tree Ensembles	622.60	-1491.269306
Capacity-Aware Beam Search	623.37	-1490.499306
Metric-Closure Facility Location	627.12	-1486.749306
Min-Max Robust Routing via Cost Perturbation	633.12	-1480.749306

As shown in Table 2 and Fig. 3, the full hybrid pipeline is necessary to achieve the minimum cost of 618.06 $/GB. When relying exclusively on the continuous LP relaxation without the heuristic ensemble fallback, the cost slightly degrades to 618.10 $/GB. More importantly, relying purely on external C-based scientific libraries (like SciPy) inside the evaluation sandbox proved highly brittle; early iterations that lacked pure-Python heuristic fallbacks crashed entirely due to multi-process module loading restrictions. The integration of a defensively programmed Randomized Shortest Path Heuristic (SPH) ensemble ensures that the system always returns a valid topology even if the LP solver encounters numerical instability or execution limits.

Conversely, when the exact mathematical solver is removed entirelyrelying only on Randomized Directed Steiner Tree Ensemblesthe cost increases significantly to 622.60 $/GB. This confirms our hypothesis that heuristic ensembles alone, while robust, fail to find the absolute minimum-cost overlapping paths in dense multi-cloud graphs.

Another critical design decision validated by our ablations is the method of fractional-to-discrete bridging. Extracting discrete routing trees from continuous LP fractions via strict randomized rounding frequently generated disconnected subgraphs.

4.4 Qualitative Analysis



(a) Pure LLM Heuristic Topology **(b) LP-based Steiner Tree Topology**

Figure 4: Qualitative comparison of multi-cloud relay trees. Panel (a) shows a sub-optimal topology generated by a pure LLM heuristic, which fails to exploit shared hops and creates redundant parallel paths. Panel (b) demonstrates the cost-optimal Steiner tree constructed by our fractional multi-commodity flow LP, which minimizes total cost by aggregating flows along a shared trunk.

To provide deeper intuition into the performance gains, Fig. 4 illustrates a qualitative comparison between topologies generated by a pure LLM heuristic and our hybrid solver. Pure heuristic approaches tend to construct independent shortest paths from the source to each destination. While this minimizes latency for individual paths, it incurs redundant egress costs when data is transmitted multiple times across the same expensive inter-cloud links. In contrast, our Fractional Multi-Commodity Flow LP formulation generates topologies that aggressively aggregate flows along shared trunks within low-cost regions (e.g., intra-AWS) before branching out to destinations in external clouds, thereby drastically reducing the total data volume subjected to high cross-cloud egress tariffs.

5 Discussion and Conclusion

In this work, we presented a hybrid neuro-symbolic routing architecture for optimizing multi-cloud data transfer topologies. By bridging the global exploration capabilities of LLMs with the rigorous local optimization of a Fractional Multi-Commodity Flow Linear Programming (LP) relaxation, our method effectively minimizes egress costs across heterogeneous cloud environments. Our approach achieves a highly competitive transfer cost of 618.06 $/GB, outperforming both human-designed heuristics [Liu et al., 2025b] and pure agentic LLM frameworks such as Engram [Karimi et al., 2026]. The integration of a randomized Shortest Path Heuristic (SPH) ensemble ensures robustness, demonstrating that continuous mathematical bounds can be successfully translated into discrete routing trees even in highly constrained search spaces.

Despite these improvements, our approach has several notable limitations. First, the reliance on exact mathematical solvers introduces significant computational overhead. During our evolutionary discovery process, early iterations strictly utilizing the continuous LP solver frequently encountered multi-process sandbox instability and timeouts. While our fallback SPH ensemble mitigates catastrophic failures, scaling the LP formulation to massive multi-cloud deployments remains computationally prohibitive. Second, the current objective function completely ignores edge throughput constraints (Gbps). Consequently, the solver may select cost-optimal but severely bottlenecked links, potentially degrading the quality of service for latency-sensitive applications [Shi et al., 2025]. Finally, our evaluation is conducted on five static network configurations, which may not fully capture the volatility of real-world cloud environments characterized by transient link failures and fluctuating spot instance pricing [AlQiam et al., 2024].

Future research should address these limitations by integrating throughput-aware edge pruning to explicitly balance financial cost with broadcast latency [Tian et al., 2025]. Additionally, replacing the monolithic LP solver with faster, distributed approximation algorithms or staged quality-diversity methods [Wei et al., 2024] could reduce computational overhead and prevent timeouts on large-scale sub-trees. Finally, extending the hybrid neuro-symbolic framework to adapt to dynamic, real-time pricing models and dynamic network topologies [Hu et al., 2026] will be crucial for deploying these automated system optimizations in production environments.

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Reproducibility Statement

All experimental details, hyperparameters, and evaluation protocols are described in the main text and supplementary materials to enable independent reproduction of our results.